MIT SCHOOL OF COMPUTING
Rajbaug, Loni-Kalbhor, Pune

MIT-ADT UNIVERSITY
PUNE, INDIA
A Leap Towards World Class Education

Name: PRANAV KOLHE Class: SOC - 08 Batch: B
Subject: Python [P.L] Roll No.: 27 Exp. No.:
Name of the Experiment:

Marks	Teacher's Signature with date
10	

Performed on:
Submitted on:

Q.1. List the key features and advantages of python. Also mention any two python IDES or editors used for program development.

→ Key features of python :—

i Python is a simple and easy-to-learn programming language.

ii It is an interpreted language, so code is executed line by line.

iii Python is high level language, which makes programming easier.

iv It supports object-oriented programming concepts.

v Python is platform independent [portable]

vi It is open-source language and free to use.

vii Python has a large standard library.

Incomplete for Theory ☐ Diagrams ☐ Observation Table ☐
Calculations ☐ Graphs ☐ Results ☐ Conclusion ☐
Understanding Through Q/A ☐ Late Submission ☐ Neatness ☐

Teacher's Signature

Advantages of python :—

i Requires less code compared to other language.
ii Easy debugging and maintainance.
iii Used in web development, data science, AI and automation.
iv Suitable for beginners as well as professionals.

Any 2 python IDES / Editors :—

i pycharm
ii Visual studio code [V.S]

2.2. Explain the history and evolution of python. Describe why python has become popular for beginners and Industry application.

→ History of Python :—

Python was developed by Guido Van Rossum in 1989 and released in 1991.
The name "python" was inspired by the comedy show Monty python's Flying circus.

Evolution of python :—

python 3.x : Improved version with better performance, security & unicode support

python 2.x : Older version.

MIT-A
UNIVERSI
PUNE, INDIA
A Leap Towards World Class Educa

Why python is popular :——

i Easy syntax similar to English.
ii Widely used in industries
iii Large community support.
iv Used by companies like Google, Netflix, and Instagram.
v Supports multiple domains such as AI, ML, used dWeb development, and cybersecurity.

Q.3. Write a python program to accept two numbers and perform arithmetic operations.

→ program :——

```
a = Float (input ("Enter First Number"))
b = Float (input ("Enter second Number"))
print ("Addition = " a+b)
print ("Subtraction = " a-b)
print ("Multiplication = " a * b)
print ("Division = " a/b)
```

Explanation :——

The program accepts two num values from the user and performs addition, substraction, Multiplication and division.

Q.4. Analyze the python code and identify data types. Explain the output.

→

```
a = 10
b = 5.5
c = "python"
d = true
```

Data Types

a ⇒ Integer

b ⇒ Float

c ⇒ string

d ⇒ Boolean.

Explaination :—

Python automatically assigns data type based on the value stored in the variable.

Q.5. Evaluate relational, logical, and bitwise operators with examples where each type of operator is most appropriate.

→ Relational Operators :—

Used to compare values code :—

```
a = 10
b = 5
print (a > b)
```

Used in decision - making statement.

Logical Operators :—
Used to combine multiple conditions
code : —

```
x = True
y = False
print (x and y)
```

Used in conditional logic

Bitwise operators :—
Used in bit-level operations.

```
a = 5
b = 3
print (a & b)
```

Used in low-level programming.

Q.6. Design and implement a menu-driven python program that.
i Accepts user input
ii performs arithmetic, relational, logic and assignment operations.
iii Displays results clearly.
The program should handle int, float. and Boolean data types.

→ code :—

```
a = int(input("Enter First number:"))
b = int(input("Enter second number:"))
```

Signature

MIT-AI
UNIVERS
PUNE, IND
A Leap Towards World Class

```
print("1. Arithmetic operation")
print("2. Relational operations")
print("3. Logical operations")
print("4. Assignment operations")

choice = int(input("Enter your choice :"))

if choice==1:
    print("Addition =", a+b)
elif choice==2:
    print("a>b =", a>b)
elif choice==3:
    print((a>0) and (b>0))
elif choice==4:
    a += b
print("result after assignment:", a)
else:
print("Invalid choice")
```

* LAB ASSIGNMENT *

Q.7. Write a program in python to display "Hello Wo

→ Algorithm : (1) start
(2) Print "Hello World"
(3) stop

MIT SCHOOL OF COMPUTING
Rajbaug, Loni-Kalbhor, Pune

MIT-ADT UNIVERSITY
PUNE, INDIA
A Leap Towards World Class Education

Name: PRANAV VITHAL KOLHE Class: SOC-08 Batch: B

Subject: PL Roll No.: 27 Exp. No.:

Name of the Experiment:

Performed on:

Submitted on:

Marks	Teacher's Signature with date
8	
10	

Q.1. List the python sequence data types and state one key characteristic of each with respect to mutability of ordering

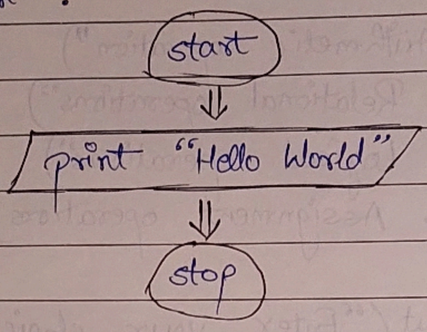
→ In python, sequence data types store collections of items in a specific order.

1. List :→ Ordered sequence
   → Mutable [elements can be changed after creation]
2. Tuple :→ Ordered sequence
   → Immutable [cannot be created once created]
3. String :→ Ordered sequence of characters
   → Immutable [characters cannot be modified]
4. Range :→ Ordered sequence of numbers
   → Immutable [cannot modify values, only generate them]

Incomplete for Theory ☐ Diagrams ☐ Observation Table ☐
Calculations ☐ Graphs ☐ Results ☐ Conclusion ☐
Understanding Through Q/A ☐ Late Submission ☐ Neatness ☐

Teacher's Signature

Flow chart :



python code :

```
print ("Hello World")
```

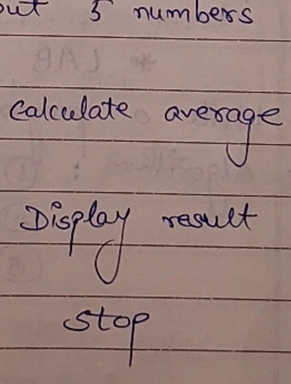
Output : Hello World

Q.8. Write a program to perform average of five numbers using input( ) function and type conversion concept :

→ Algorithm : —

1 start
2 Input five numbers
3 Convert to float
4 Calculate average
5 Display result
6 stop

Flow chart : —



code :—

```
a = float(input("Enter numbers 1 :"))
b = float(input("Enter number 2 :"))
c = float(input("Enter number 3 :"))
d = float(input("Enter number 4 :"))
e = float(input("Enter number 5 :"))

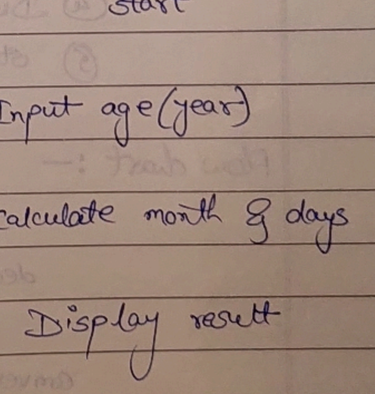
avg = (a+b+c+d+e)/5
print("Average =", avg)
```

Q.9. Write a program that prompts the user to enter their age and prints out their age in years, month, days.

→ Algorithm :—

1. Start
2. Input age in years
3. convert to month & days
4. Display result
5. stop

Flow chart :—



MIT UNIVERSITY
ART, DESIGN & TECHNOLOGY
MIT UNIVERSITY
PUNE, INDIA
A Leap Towards World Class

python code :—

```
age = int (input ("Enter your age years:"))

months = age * 12
days = age * 365

print ("Age in years:", age)
print ("Age in month :", month)
print ("Age in day :", days)
```

Q.10. Write a program to demonstrate the concept of data types and conversion

→ Algorithm : ① start
② Declare variables of different data types
③ Convert data types
④ Display values
⑤ stop

Flow chart :— start

declare variable

convert data types

Display results

stop



python code : —

```
a = 10
b = 5.5
c = "20"

print (type (a))
print (type (b))
print (type (c))

c = int(c)
print ("After conversion :", c)
print (type (c))
```